

ALMOST EQUIVALENT: MEASURING WHEN DOWNSCALED IMAGES CARRY THE FLOW-MATCHING TRAINING SIGNAL

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ABSTRACT

Scale-wise diffusion schemes rest on a spectral argument: flow-matching noising progressively masks high spatial frequencies, so above some noise level a down-scaled latent carries *almost* the full surviving signal. We ask whether this licenses a training-side substitution — can the gradient of a low-rank adapter be computed from a downscaled latent at high noise without changing what is learned? We first derive what the substitution can cost: a first-order account in which the gradient gap decomposes into a noise-dependent *input term*, read through the gradient norm, plus a σ -independent *floor* carried by the network’s input–output Jacobian — and we embed the spectral argument, in its strongest tolerance-parameterized form (Xiao et al., 2026), as the null model of the input term. On a measured latent spectrum the null is structurally confined: a one-parameter family of per-route boundaries that depends on the downscale *ratio* alone, is compressed to a ~ 0.13 spread in σ , and admits no floor. Per-noise-level gradient measurements on a DiT-based flow-matching model, with a fresh-noise floor and a VAE re-encoding control, reject the null on every axis where the accounts diverge: the mildest route’s crossover sits at $\sigma \approx 0.5$ and no tolerance reproduces the map (the one matching it predicts two never-safe routes safe by $\sigma = 0.63$); a $2\times$ downscale is never gradient-safe at any noise level, retaining a gap of 0.3 at $\sigma = 0.94$ where the input is 97% noise by power; and the floor is governed by the *absolute token count* of the downscaled grid, concentrates in early blocks, and separates — by a positional-interpolation intervention — into a rotary-position-embedding share and an irreducible graph residue. The reason is structural: expected gradients average noise out rather than being masked by it, and the flow-matching target $v = \epsilon - x$ carries the clean image at unit amplitude at every σ . The practical output is a measured (route, σ) safety map and an opt-in σ -conditional trainer that draws the noise level first and substitutes the downscaled latent only on measured-safe steps, for a $\sim 14\%$ wall-clock reduction at fixed step count on our corpus.

1 INTRODUCTION

The cost of a diffusion-transformer forward pass grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. A family of recent methods therefore runs part of the computation at reduced resolution, justified by a spectral observation about the forward noising process: under flow-matching noising $z_\sigma = (1 - \sigma)x + \sigma\epsilon$, the per-frequency signal-to-noise ratio falls with σ , and high spatial frequencies drop below the noise floor first. Above a crossover noise level, a downscaled latent is claimed to carry *almost* the full surviving signal — the premise behind scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), pyramidal flow matching (Jin et al., 2024), and progressive-resolution training schedules more broadly (Karras et al., 2018; Chen et al., 2024; Hoogeboom et al., 2023).

These methods change the *inference* trajectory or the model itself. We ask a narrower, training-side question: at high noise, can the *gradient* of an adapter be computed on a downscaled latent as a drop-in substitute for the native one — same model, same objective, same σ distribution, only the spatial grid changed on some steps? If the spectral account governed, the answer would be yes above a per-route boundary — for our latents, $\sigma \approx 0.14$ at the classical equal-power tolerance, and $\sigma \approx 0.6$

even at the strict tolerance SPD ships (§3.3). The payoff would be large: on our corpus roughly half of all training draws sit at $\sigma > 0.5$.

Rather than measuring first and rationalizing after, we first derive what the substitution can cost, and then let the measurements discriminate. The contributions:

1. **A two-term account of the demotion gap, derived before it is fit** (§3). Writing the adapter gradient as $g = J^T r$ (Jacobian times residual), demotion perturbs exactly two things — the information the input still carries, and the compute graph itself — and first-order perturbation plus cosine geometry force the form $\text{gap}_e(\sigma) \approx A_e m(\sigma)/G(\sigma)^p + \text{Floor}_e$ with $p \in [1, 2]$: a noise-dependent input branch read through the U-shaped gradient norm, plus a σ -independent floor carried by the network function. The derivation fixes the *form*; the coefficients are measured, and we keep an explicit ledger of what is derived, what is assumed, and what is measured. We then embed the spectral argument, in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026), as the null model of the input branch — and show, before any gradient is measured, that on our latent spectrum the null is a one-parameter family that depends on the downscale *ratio* alone, is compressed to a ~ 0.13 spread in σ across all tolerances, and structurally admits no floor.
2. **Measurements that reject the null on every axis where the accounts diverge** (§4). Our instrument computes, per noise-level bin, the cosine between adapter gradients from native and downsampled latents, calibrated by a fresh-noise floor and a VAE re-encoding control. The measured crossover for the mildest downscale ($0.875\times$ edge) is $\sigma \approx 0.5$; no tolerance reproduces the map (the one matching this route predicts the two never-safe routes safe by $\sigma = 0.63$); a $2\times$ downscale never becomes safe, retaining a gap of ~ 0.3 at $\sigma = 0.94$ where the input is essentially pure noise. Term by term: the floor is real, σ -independent, governed by absolute target token count rather than ratio, concentrated in early blocks, and decomposes — via an exact positional-interpolation intervention on the rotary embedding — into a coordinate-system share RoPE_e and a non-positional graph residue Resid_e ; the input branch’s amplitude is governed by the downscale ratio (an iso-severity probe at $1.6\times$ different absolute size reproduces the same curve); and the prediction-residual mismatch is *route-uniform*, placing all route identity in the Jacobian — the residual-level kill of the diagonal spectral null.
3. **A measured safety map and a practical trainer** (§5). Safety is a per-route boundary $\sigma^*(\text{route})$, not a universal ratio rule: $\{1024 \rightarrow 896 \text{ at } \sigma > 0.5; 1280 \rightarrow 1024 \text{ at } \sigma \gtrsim 0.75\}$, while $896 \rightarrow 768$ and every $\rightarrow 512$ route are never safe. We wire the measured-safe route into an opt-in σ -conditional trainer — draw σ first, fetch the downsampled latent only when the whole batch is above the gate — for a $\sim 14\%$ wall-clock reduction at a fixed step budget, with a σ -gated banded rotary alignment as a measured refinement.

The presentation is theory-first; the practice was observational-first: verdict criteria were frozen before each probe ran, the two-term account was pre-registered with an outcome matrix (including a kill switch) before the discriminating runs, and each mechanistic claim is grounded in a designated falsification test (several of which fired and killed proposed routes; we report those too).

2 RELATED WORK

Scale-wise and progressive-resolution diffusion. Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogetboom et al., 2023) train at increasing resolution as a schedule. Pyramidal flow matching (Jin et al., 2024) and scale-wise distillation (SwD) (Starodubcev et al., 2025) tie resolution to the noise level itself, with SwD giving the spectral analysis we take as the hypothesis under test: their Fig. 1-style argument shows the noised downsampled latent is close in distribution to a downsampled noised latent above a crossover σ . Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives that argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model — a criterion on the *Bayes-optimal predictor’s* per-band output, not merely on input SNR — from which an inference-time resolution schedule follows; we adopt their Proposition 1 as the null model of our input branch (§3.3). SPD also fine-tunes (with LoRA) against *stage-specific modified velocity targets* along the progressive trajectory — changing the objective per resolution stage rather than substituting the latent under

the native objective, which is our question; that they observe a training–inference gap motivating this fine-tuning is independently consistent with our finding that resolution changes are not free at the training level. Our contribution is orthogonal to all of these: we measure whether input-level closeness transfers to the *training gradient*, holding model and objective fixed — and find it does not transfer at any tolerance.

Flow matching and DiT. We work in the flow-matching / rectified-flow formulation (Lipman et al., 2023; Liu et al., 2023; Esser et al., 2024) on a Diffusion Transformer (Peebles & Xie, 2023) with rotary position embeddings (Su et al., 2024). Adapters are LoRA (Hu et al., 2022).

Positional extension. Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* — matching the downscaled grid’s relative phase geometry to native — to decompose the resolution-sensitivity floor, and adopt the σ -gated band schedule of Zhao et al. (2026) (functional form only; their spectral scale laws mispredict our measured boundaries, consistent with our main finding) as a training-time refinement.

Gradient-agreement methodology. Reading training decisions off gradient cosines against a noise-redraw floor follows the general logic of gradient-noise-scale analyses (McCandlish et al., 2018); our estimator operates per σ -bin with means over images, the regime we found reliable (split-half reliability 0.73–0.88), rather than per-image rankings, which were pure noise in our earlier attempts.

3 WHAT DEMOTION CAN COST: A TWO-TERM ACCOUNT

Objective and noising. With clean latent x , noise $\epsilon \sim \mathcal{N}(0, I)$, and $\sigma \in (0, 1]$, the noised input is $z_\sigma = (1 - \sigma)x + \sigma\epsilon$ and the flow-matching target is $v = \epsilon - x$; the loss is $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$ with gradients taken with respect to adapter parameters only.

Demotion. A *demotion* $e_{\text{nat}} \rightarrow e$ replaces the native latent with one produced by pixel-space downscaling of the source image to the tier- e bucket followed by VAE re-encoding — the ordering SwD validated (“strategy B”); we never downsample in latent space. The caption, the noise draw structure, and the σ under test are held fixed; only the spatial grid changes.

The verdict quantity. For a set of images and a σ -bin, adapter gradients are accumulated per arm and compared by cosine; the *demotion gap* is

$$\text{gap}_e(\sigma) = \cos_{\text{floor}}(\sigma) - \cos(g_{\text{nat}}(\sigma), g_e(\sigma)), \quad (1)$$

where $\cos_{\text{floor}}$ is the same-bin cosine between two independent native draw sets — the “how much do gradients differ anyway” null. The full instrument, including a VAE re-encoding control that prices the round trip demotion necessarily pays, is specified in §4.1; this section asks what structure Eq. 1 can have.

3.1 WHY INPUT-LEVEL SPECTRAL SUFFICIENCY CANNOT BOUND THE GRADIENT

The spectral argument is a statement about the *input*. Two observations, neither involving the spectrum of z_σ , show it cannot bound the training gradient:

(a) Expected gradients are never noise-masked. “Noise power exceeds signal power in band f ” is a statement about a single sample of z_σ . The quantity training consumes is $\mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$ — an expectation that averages the noise out and retains attenuated signal at *any* SNR. Input-level sufficiency licenses inference-time substitution; it never licenses training-dynamics equivalence.

(b) The target carries the clean image at unit amplitude at every σ . The flow-matching target is $v = \epsilon - x$. The *input* is progressively noise-masked as $\sigma \rightarrow 1$, but x sits in the *target* at coefficient one even at $\sigma = 0.94$. Writing the adapter gradient as $g = J^\top r$ (input–output Jacobian times prediction residual), the two grids can disagree through J itself — attention softmax over a different token count

N , rotary phase density, sequence-length-dependent normalization — regardless of how little image signal the input retains.

Together these say a demoted input can disagree with the native gradient through two distinct channels: what the input still *carries* (information), and what the network *does* with a different grid (computation). The rest of this section turns that observation into a measurable form.

3.2 THE FIRST-ORDER FORM

Demotion changes exactly two things about the map from (image, noise, caption) to adapter gradient:

1. **the input branch:** spatial-frequency bands above the coarse grid’s Nyquist limit are absent from input *and* target (and the re-encode perturbs what survives) — a perturbation $\Delta r(\sigma)$ of the prediction residual whose size is governed by posterior physics and shrinks as the model’s reliance on input detail shrinks with σ ;
2. **the graph:** token count, rotary phase density, sequence-length-dependent normalization — an operator perturbation ΔJ that is σ -*independent*, because the grid change does not reference σ .

To first order the gradient perturbation is $\delta g \approx J^\top \Delta r(\sigma) + \Delta J^\top r$, with the cross term second-order. One caveat is worth stating precisely: J_{nat} and J_e act on different grids, so the two terms are not literal matrix products across arms; they are defined *operationally*, as the two independent perturbation directions in the shared adapter-parameter space where both gradients live. This is exactly what the designated probes of §4.3 isolate: the $\sigma=1$ endpoint makes the input branch vanish in distribution (the input is pure ϵ on every grid), and the x-zero probe removes the image from input and target entirely — each isolating the graph term by construction.

Cosine geometry then fixes how a mismatch is read. For a perturbation δg with component $\delta g_\perp$ orthogonal to g ,

$$1 - \cos(g, g + \delta g) \approx \frac{1}{2} \|\delta g_\perp\|^2 / \|g\|^2 \quad (2)$$

in the small-mismatch limit, degrading toward $\|\delta g_\perp\|/\|g\|$ when the mismatch is comparable to the signal. Any absolute mismatch is therefore read through the total gradient scale $G(\sigma) = \|g\|$, with an exponent pinned to $p \in [1, 2]$ by the geometry alone — 2 in the small-orthogonal-mismatch limit. Under three assumptions — **(A1)** the two perturbations are uncorrelated in parameter space (no interaction term), **(A2)** the graph term’s *relative* footprint $\|\Delta J^\top r\|/\|g\|$ is σ -stationary (its numerator and denominator share the residual’s scale), and **(A3)** the input branch’s gain through $J^\top$ is σ -independent, so it factorizes into a route amplitude times a σ -shape — the gap takes the two-term form

$$\text{gap}_e(\sigma) \approx \underbrace{A_e \cdot m(\sigma)}_{\text{input branch } S_1} / \underbrace{G(\sigma)^p}_{\text{graph term}} + \underbrace{\text{Floor}_e}_{\text{graph term}}, \quad p \in [1, 2], \quad (3)$$

where $m(\sigma) = \|\Delta \bar{r}(\sigma)\|$ is the mismatch of the model’s mean prediction residual across grids (directly measurable, §4.4), A_e is a σ -independent route amplitude absorbing units and the Jacobian gain, and Floor_e is the σ -independent term carried by the network function itself.

The ledger, explicitly: the *form* — additive two-term structure, the G^p denominator, $p \in [1, 2]$, the σ -flatness of the floor — is derived from the perturbation expansion plus cosine geometry, conditional on A1–A3; the assumptions are tested by designated probes (§4.3); the coefficients A_e , $m(\sigma)$, $G(\sigma)$, Floor_e are measured, not derived. The account was pre-registered with an outcome matrix (including a kill switch: all endpoint gaps ≈ 0 would have falsified the graph term) before the discriminating runs.

3.3 THE SPECTRAL ACCOUNT AS A NULL MODEL OF THE INPUT BRANCH

The spectral argument, made precise, is a model of the input branch alone — and its strongest available form is SPD’s tolerance-parameterized crossover (Xiao et al., 2026). Under the modeling assumption $x_0^{(\omega)} \sim \mathcal{N}(0, P_\omega)$ per frequency ω (with P the clean-latent power spectrum), their Proposition 1 states that for all $\sigma \geq t_\omega$ the *Bayes-optimal* per-band velocity prediction is within tolerance

δ of pure noise, $\mathbb{E}[|v^{*(\omega)} - \epsilon^{(\omega)}|^2] \leq \delta$, at

$$t_\omega = \frac{1}{1 + \sqrt{\delta / (P_\omega (1 + P_\omega - \delta))}}, \quad (4)$$

and their Proposition 2 places the safe-substitution boundary for a coarse grid at the t_ω of its Nyquist band. This is a prediction-level criterion, strictly stronger than raw input SNR; the often-quoted equal-power crossover $\sigma_{\text{eq}}(f) = \sqrt{P(f)/(1 + \sqrt{P(f)})}$ is its $\delta = (1 + P_\omega)/2$ slice.

Taken as the null model of S_1 , Eq. 4 has three structural properties, each checkable before any gradient is measured:

(N1) One-parameter family. All route boundaries move together under the single tolerance δ ; the family is totally ordered.

(N2) Ratio-only. In normalized frequency the demoted grid’s Nyquist sits at $f_{\text{cut}} = \frac{1}{2} e/e_{\text{nat}}$ — a function of the downscale *ratio* alone. The null therefore cannot distinguish routes of equal ratio at different absolute sizes, and assigns nearly identical boundaries to nearly equal ratios regardless of grid size. It is structurally blind to any absolute-size effect.

(N3) No floor. For every route and every δ , the null’s gap vanishes above a finite boundary: every route becomes safe at some σ .

And under the same diagonal-Gaussian assumption the null extends to the residual level: the posterior mean is a per-band Wiener shrinkage, so with no cross-frequency coupling the cross-grid mean-residual mismatch is confined to the destroyed band and **(N4)** ordered across routes by destroyed-band energy.

Table 1 instantiates the family on our measured latent spectrum (§4.2; P at the 896/768/512 cuts = 0.025/0.029/0.065). Because the latents are high-frequency-quiet, P barely varies across the cuts and the family is *compressed*: the spread between the mildest and harshest route never exceeds ~ 0.13 in σ at any tolerance. In particular the two routes our measurements will separate maximally — 1024→896 (ratio 0.875) and 896→768 (ratio 0.857) — receive boundaries within 0.01 of each other at *every* δ .

Which tolerance? Eq. 4 needs a δ , and it is worth asking where one could legitimately come from. SPD’s shipped $\delta = 0.01$ is an inference-side choice, selected on a speed–quality Pareto ablation over generated images (Xiao et al., 2026) — nothing ties it to gradients. Machine precision is not the relevant floor either: bf16 rounding on unit-scale per-band amplitudes injects power of order 10^{-5} or below, orders of magnitude beneath any boundary-moving tolerance. Our setting, however, fixes a tolerance *by construction*: the demotion recipe pays a resize–VAE-encode round trip whose latent error has measurable per-band power $D(f)$ — the same pipeline noise the re-encoding control prices at the gradient level — and D is expressed in exactly Eq. 4’s units (per-band power against unit noise). Band content below $D(f_{\text{cut}})$ is invisible to the substitution pipeline even with no resolution change, so $\delta_{\text{reenc}}(e) := D(f_{\text{cut}}(e))$ anchors the null with zero free parameters. On the probe set this floor and its implied boundaries are **[pending: reenc_noise_floor.py, queued]**; we will report the anchored row alongside the family in Table 1. Either way the confrontation below does not hinge on the choice: the structural failures (N2, N3) are δ -independent.

3.4 THE GRAPH TERM: A POSITIONAL SHARE AND A GRAPH RESIDUE

The operator perturbation ΔJ has identifiable parts. Changing the grid changes (i) the rotary coordinate system — every band of the position embedding accumulates phase at a different density across the image — and (ii) non-positional graph statistics: attention softmax over a different token count, sequence-length-dependent normalization, and the sheer capacity of the coarse graph to approximate the fine graph’s computation. Splitting $\text{Floor}_e = \text{RoPE}_e + \text{Resid}_e$ yields one sharp prediction: the positional share is a *coordinate-system artifact*, so an exact positional-interpolation intervention — evaluating the demoted grid at fractional rotary coordinates that reproduce the native relative phase geometry — should remove RoPE_e wherever the input branch is inactive ($\sigma \rightarrow 1$), and a band-counting sketch says the residue should grow with absolute grid change (the number of rotary bands whose per-extent rotation count crosses decorrelation grows with the extent), attaching

Table 1: The spectral null (Eq. 4) instantiated on the measured latent spectrum: predicted safe boundaries t^* per route as the tolerance δ sweeps its range. SPD ships $\delta = 0.01$; the equal-power slice is the classical crossover. No row reproduces the measured pattern (last line): the δ that matches the safe route’s boundary predicts both never-safe routes safe by $\sigma = 0.63$.

tolerance δ	1024→896	1024→768	1024→512	
$(1 + P_\omega)/2$ (equal power)	0.14	0.15	0.20	classical crossover
0.05	0.41	0.43	0.53	
0.025	0.50	0.52	0.62	tuned to the measured 896 gate
0.01	0.61	0.63	0.72	SPD’s shipped default
0.005	0.69	0.71	0.79	
measured (§5.1)	≈ 0.5	never safe	never safe	floors 0 / 0.06–0.13 / ~ 0.3

the absolute-size governor to Resid_e . We flag the sketch’s epistemic status plainly: like YaRN’s own band thresholds (Peng et al., 2024), its constants are calibrated, not derived; we claim no closed form for Floor_e .

3.5 WHERE THE ACCOUNTS DIVERGE

The two-term account (Eq. 3) and the spectral null (Eq. 4) agree that gaps shrink with σ and order by severity. They diverge on five measurable axes, which the experiments of §4 score:

1. **Boundary pattern:** the null predicts some row of Table 1; the account permits {finite boundary, never, never} if floors differ (§4.2).
2. **High- σ persistence:** the null predicts every gap vanishes as $\sigma \rightarrow 1$; the account predicts persistence equal to Floor_e , surviving even pure-noise input (§4.3).
3. **Governors:** the null is ratio-only (N2); the account permits ratio to govern A_e while absolute size governs Floor_e (§4.5).
4. **Residual ordering:** the null predicts route-ordered residual mismatch (N4); if route identity instead lives in J , the residual curves are route-uniform under route-ordered gaps (§4.4).
5. **Curve shape:** the null ties the gap’s shape to per-band SNR; the account ties it to $m(\sigma)/G(\sigma)^p$ — an interior peak where the U-shaped G is smallest, and a low- σ dip that renormalizing by G removes (§4.4).

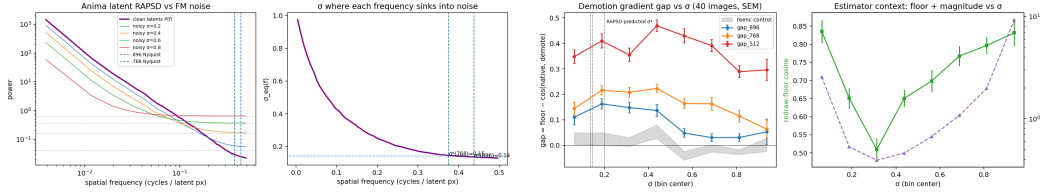
4 EXPERIMENTS

4.1 INSTRUMENT

Model and data. All measurements use Anima, an open DiT-based flow-matching text-to-image model (28 transformer blocks, 3D rotary position embeddings, Qwen-family VAE), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length $e \in \{512, 768, 896, 1024, 1280\}$; a tier- e image occupies roughly $(e/16)^2$ latent-patch tokens (e.g. $\sim 4,100$ tokens at 1024, $\sim 3,000$ at 896, $\sim 2,160$ at 768, $\sim 1,000$ at 512). The probe adapter is a LoRA soup checkpoint trained at native tiers; we return to this operating-point caveat in §6.

The gradient probe. For each image and each of B uniform σ -bins we accumulate adapter gradients over D stratified noise draws per arm (shipped runs: 8×8 unless noted; $N = 40$ images) and compare arms by cosine similarity of the flattened accumulated gradients (Eq. 1):

- **floor:** two independent draw sets at native resolution — the redraw null;
- **reenc:** decode–re-encode at native resolution — isolates the VAE round-trip cost that demotion necessarily pays;
- **demote arms:** one per edge e under test.



(a) Latent RAPSD and the equal-power slice $\sigma_{eq}(f)$ of the spectral family. Predicted σ^* : 0.136 / 0.146 / ~ 0.20 times, $N=40$). The 896 route floors at $\sigma \approx 0.5$; 512 never for demotion to 896 / 768 / 512. (b) Measured demotion gap per σ -bin (1024-tier natives, $N=40$). The 896 route floors at $\sigma \approx 0.5$; 512 never for demotion to 896 / 768 / 512.

Figure 1: The spectral family (left, equal-power slice) versus the gradient measurement (right). No tolerance reproduces the measured pattern (Table 1): the equal-power crossover is off by $\sim 3.5\times$ and the $2\times$ downscale never becomes safe.

Table 2: Demotion gap by σ -bin, 1024-tier natives ($N=40$, bin-mean SEM ~ 0.02 , re-encoding control $|gap_{reenc}| \leq 0.054$ everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. The equal-power slice of the spectral family predicts collapse above the second bin for every route; no other slice fits either (Table 1).

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
gap ₈₉₆ (ratio 0.875)	.110	.162	.148	.137	.048	.030	.030	.053
gap ₇₆₈ (ratio 0.75)	.144	.216	.208	.223	.164	.163	.115	.063
gap ₅₁₂ (ratio 0.5)	.348	.410	.355	.469	.430	.391	.289	.296
cos _{floor}	.84	.65	.51	.65	.70	.77	.80	.83
$\ g\ $ (native)	2.6	0.5	0.4	0.5	0.7	1.1	2.0	9.3

The gap is in cosine units (a mismatch *fraction*, scale-free; bin-mean SEM ~ 0.02 at $N=40$). Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with $\|g\|$ across bins. The re-encoding control $gap_{reenc} \approx 0$ (band ± 0.04) is the instrument-validity witness in every run, and every bin-mean curve must pass a split-half reliability check over images — a discipline forced on us by an earlier failure in which per-image rankings from the same gradients had split-half reliability indistinguishable from zero. Uniform σ -bins make the training-time σ -density irrelevant to the map; density-weighting the bins reproduces our earlier pooled (marginalized) measurements as a consistency check. Estimator details and the endpoint / x-zero probe modes are specified in Appendix A.

4.2 THE GAP CURVES VERSUS THE SPECTRAL FAMILY

The spectrum. Anima’s VAE latents are high-frequency-quiet ($P(f) < 1$ above $f \approx 0.16$ cycles/latent-pixel; latent variance 0.42), giving the equal-power crossovers $\sigma^*(896) = 0.136$, $\sigma^*(768) = 0.146$, $\sigma^*(512) \approx 0.20$, with tight per-image spread (p10–p90 ≈ 0.11 – 0.16): the spectral crossover is image-generic, and at that tolerance demotion would be safe over almost the entire σ axis (Fig. 1a). The full tolerance family these cuts admit is Table 1.

The measurement. Table 2 gives the measured gap curves for 1024-tier natives demoted to 896/768/512.

Three reads, against criteria frozen before the run: (i) σ -dependence is real and in the predicted direction (Spearman $\sigma \rightarrow gap$: $-0.69/-0.57/-0.36$), and tier ordering holds at every bin — the *qualitative* spectral intuition survives; (ii) the crossover for the mildest route is $\sigma \approx 0.5$ — $3.5\times$ the equal-power prediction, and no tolerance rescues the family: the δ that matches this boundary ($\delta = 0.025$, Table 1) predicts 768 safe at 0.52 and 512 safe at 0.62, while SPD’s own shipped $\delta = 0.01$ predicts all three routes safe by 0.72; (iii) the $2\times$ downscale route retains gap ≈ 0.3 at $\sigma = 0.94$, where the input is $\sim 97\%$ noise by power in every band the coarse grid cannot represent — the high- σ persistence of axis 2 in §3.5, which no member of the null family can produce. There is no noise level at which a $2\times$ -downscaled latent yields the native gradient.

Table 3: Isolating the floor. **Endpoint:** at $\sigma=1$ the input is exactly ϵ , so any gap *is* the floor by construction (16 draws). **x-zero:** the image is zeroed in both input and target on every grid; any surviving gap is pure graph-shape sensitivity. Pre-registered bands in brackets; all pass, and the account’s kill switch (all endpoint gaps ≈ 0) does not fire.

route	endpoint gap @ $\sigma=1$ (SEM)	band	x-zero @ $\sigma=1$ (SEM)
re-encoding control	−0.038 (.032)	[−.04, .04]	—
1024→896	−0.009 (.042)	[0, 0.08]	0.004 (.018)
1024→768	+0.127 (.054)	[0.04, 0.15]	0.064 (.028)
1024→512	+0.326 (.059)	[0.2, 0.4]	0.299 (.040)

Ratio does not transfer — the null’s blind spot, measured. Re-running the probe one tier down (896-tier natives, demote to 768/512) fails its pre-registered bar: 896→768 (ratio 0.857, nearly identical to the passing 0.875) retains a high- σ residual of 0.06–0.12, roughly 2 $\times$ the 1024→896 plateau and outside the re-encoding band, while the two-tiers-down control stays elevated everywhere (Appendix Table 6). This is exactly property N2’s blindness made concrete: the null assigns these two routes boundaries within 0.01 of each other at every tolerance, and the measurement separates them into *safe* versus *never safe*. “One tier down at high σ ” is not the invariant; §4.5 locates why (the floor is governed by absolute size, and 896→768’s floor alone exceeds the band).

4.3 THE FLOOR EXISTS, IS σ -INDEPENDENT, AND IS NOT CARRIED BY THE INPUT

Two designated probes (Table 3) establish the graph term — and with it assumptions A1–A2 of §3.2. At the $\sigma = 1$ *endpoint* the input is pure ϵ — identical in distribution across arms — so the input-information term is zero by construction, and the measured gaps ($\approx 0 / 0.13 / 0.33$ for 896/768/512) match the high- σ plateaus of Table 2: the plateau *is* the floor, exactly the additivity the form requires. The *x-zero* probe removes the image from input *and* target everywhere; for the 2 $\times$ route the surviving gap equals the endpoint gap (0.30 vs 0.33) — the floor is essentially all network-function term, with per-image content contributing nothing. Two further exonerations: the re-encoding control sits in band (the VAE round trip is not the cause), and a forward-only probe of the model’s caption-conditioned *prior* across grids shows route distances that are flat where the floors are strongly ordered — so the floor’s route-ordering lives in the Jacobian factor J , not in the model’s resolution-conditioned prior $\hat{x}_{\text{prior}}$, and not in a training-distribution discontinuity (the never-trained 1280 grid is no more distant from 1024 than heavily-trained adjacent pairs are from each other).

4.4 THE INPUT BRANCH: A ROUTE-UNIFORM NUMERATOR OVER A U-SHAPED DENOMINATOR

$m(\sigma)$ is universal — the residual-level null fails. Measuring the mean prediction residual $\bar{r} = \mathbb{E}_{\epsilon} [\hat{y} - (\epsilon - x)]$ per grid and σ — the exact r in $g = J^T r$ — shows a strong, monotone σ -shape (cross-grid excess 0.89 $\rightarrow$ 0.36 in relative L_2 from $\sigma=0.125$ to 1) that is the *same* for *every* route within ± 0.02 , across routes whose gradient floors span 0 to 0.3 and whose crossovers span 0.5 to nonexistent (Appendix Table 7). This kills the diagonal null at the residual level (N4): with no cross-frequency coupling, its mismatch is confined to the destroyed band and ordered by destroyed-band energy, which differs substantially across these routes — route-uniformity says the real mismatch is carried by cross-band structure the diagonal model cannot express. All route identity therefore lives in J , not in the prediction error — a pre-registered null that would equally have falsified the two-term account had the residual curves been route-ordered. The decay is smooth, Wiener-like posterior shrinkage with no hard gate at the spectral crossover: gap curves read as “collapsed” where S_1 sinks below the instrument band, not at σ_{eq} .

$G(\sigma)$ shapes the curve. Because the gap is a cosine it measures a mismatch *fraction*; the total gradient norm is U-shaped in σ (2.6 $\rightarrow$ 0.4 at $\sigma \approx 0.3 \rightarrow 9.3$; Table 2, last row) — large at high σ where the model is pulling composition out of the prior and the residual is image-scale, small in the mid- σ refinement regime, and inflated again at low σ by irreducible ϵ . A monotone absolute mismatch divided by a U-shaped norm yields exactly the measured interior-peaked gap curves: bin-mean gap tracks $1/\|g\|$ (Spearman +0.55 to +0.90 across arms and runs, and within images), and renormalizing by G flips the anomalous low- σ dip into the monotone maximum in every arm of

Table 4: Iso-severity probe: gap by σ -bin on the 1280-tier cache ($N=24$, 4 draws/bin) against the corpus 1024 $\rightarrow$ 896 curve. The ratio-matched routes coincide within ~ 1.4 combined SEM at every bin despite a $1.6\times$ difference in absolute target size; the same-native harsher-ratio route separates. *896’s canonical 16-draw endpoint is -0.009 (Table 3).

σ	0.125	0.375	0.625	0.875	1.0
re-encoding control	+.047	−.005	+.048	+.020	−.012
1280 $\rightarrow$ 1120 (ratio .875)	.129	.110	.054	−.012	−.007
1280 $\rightarrow$ 1024 (ratio .80)	.170	.178	.111	.002	.061
1024 $\rightarrow$ 896 (ratio .875)	.132	.076	.077	.049	.100*

both runs — axis 5 of §3.5, resolved for the two-term account. This is a post-hoc consistency analysis (flagged as such): it explains the *shape*; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

4.5 THE GOVERNORS: RATIO SETS THE AMPLITUDE, ABSOLUTE SIZE SETS THE FLOOR

With the floor isolated, what orders the routes? Two probes give a clean division of labor — axis 3 of §3.5:

Absolute size, not ratio, sets the floor. The route 1280 $\rightarrow$ 1024 is *more* aggressive by ratio (0.80) than the never-safe 896 $\rightarrow$ 768 (0.857), yet its floor is ≈ 0 and it enters the re-encoding band at $\sigma \geq 0.75$ — refuting any pure ratio rule for the floor, and with it the null’s only degree of freedom. Conversely the floor grows monotonically as the *target* grid shrinks (0/0.06–0.13/ ~ 0.3 at target $\sim 3,000/2,160/1,000$ tokens), consistent with “floor = how well the coarse graph approximates the fine graph’s computation.”

Ratio, not absolute size, sets the amplitude A_e . A synthetic iso-severity route 1280 $\rightarrow$ 1120 — edge ratio 0.875 exactly matched to 1024 $\rightarrow$ 896, but $1.6\times$ its absolute target size — reproduces the 1024 $\rightarrow$ 896 curve bin-for-bin and its crossover $\sigma^* \approx 0.5$, while the same-native 1280 $\rightarrow$ 1024 (ratio 0.80) separates and floors later (Table 4). Amplitude follows ratio; with $\text{Floor}_{1120} \approx 0$ this completes the division: *ratio sets A_e* ; *absolute target capacity sets Floor_e* . The pre-registered contrast (ratio-governed $\Rightarrow \sigma^* \approx 0.5$; capacity-governed $\Rightarrow$ earlier floor) discriminated cleanly. Note what survives of the spectral account here: ratio *is* the right governor for the input branch — the null’s error is not its governor but its scope, mistaking one term for the whole gap.

4.6 WHERE THE FLOOR LIVES: DEPTH, AND A POSITIONAL DECOMPOSITION

Depth, not module type. Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) shows the floor concentrated in *depth*: early blocks (0–9, peaking at 3–8) carry $\sim 3\times$ the late-block gap, uniformly across every module type within a block (at $512/\sigma=1$, every type sits in 0.22–0.31; Appendix Table 8). The content share (endpoint minus x-zero) is a late-block minority effect. The mechanism picture: the first ~ 10 blocks build grid-calibrated token statistics; the divergence propagates into every parameter type’s gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Notably, query and key projections show *zero* excess over value projections — which we initially read as refuting a rotary-embedding mechanism. That inference was wrong, in an instructive way.

$\text{Floor}_e = \text{RoPE}_e + \text{Resid}_e$. Landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. The discriminating experiment is the origin-side intervention of §3.4: evaluate the demoted grid with positionally-interpolated fractional rotary coordinates (patch i at $i \cdot H_{\text{nat}}/H_{\text{dem}}$ per axis), making its relative phase geometry exactly match native. At the $\sigma=1$ endpoint this *erases* the 768 floor ($+0.080 \rightarrow -0.001$; paired $\Delta = +0.081$, 78% of images improved) and removes $\sim 30\%$ of the 512 floor ($+0.320 \rightarrow +0.224$), while leaving the in-band 896 control untouched (Appendix Table 9). The mild-route floor is mostly a *coordinate-system artifact*; the harsh-route bulk is a genuine non-positional graph residue (attention-over- N statistics, sequence-length-dependent normalization, capacity). The capacity governor of §4.5 attaches to Resid_e , as the band-counting sketch anticipated.

Table 5: The measured safety map. σ^* = noise level above which the demotion gap is within the re-encoding control band; token ratio = per-forward cost of the demoted grid. Every entry is a bin-mean read at $N \geq 24$ with split-half reliability and in-band controls.

route	edge ratio	target tokens	Floor _e	σ^*	verdict
1024→896	0.875	~3,000	≈ 0	≈ 0.5	safe , $\sigma > 0.5$
1280→1024	0.80	~4,100	≈ 0	$\in (0.625, 0.875)$	safe , $\sigma \gtrsim 0.75$
1280→1120	0.875	~4,800	≈ 0	≈ 0.5	safe (synthetic grid)
896→768	0.857	~2,160	0.06–0.13	—	never in band
1024→768	0.75	~2,160	0.06–0.13	—	never in band
any →512	≤ 0.57	~1,000	~0.3	—	never in band

The positional share is a noise-regime artifact only. A σ -resolved follow-up closes the tempting route this opens (training 1024→768 through interpolated coordinates): with image content in the input, the stretched forward is *off-manifold* — the interpolated arm is *worse* than the plain demoted arm through $\sigma \in [0.56, 0.81]$ (paired -0.05 to -0.11) and wins only at $\sigma \geq 0.94$. The floor’s positional share is erasable only where the input is noise-dominated, so the decomposition is a mechanism finding, not a training lever — and the input-branch term of the 768 route is fatal on its own ($+0.09$ – 0.22 over control across the window). A frequency-*selective* banded alignment does survive in-window (§5.2).

4.7 SCORING THE ACCOUNTS

Against the five axes of §3.5, the null loses each one: the boundary pattern is $\{0.5, \text{never}, \text{never}\}$ (axis 1, §4.2); gaps persist at pure-noise input, equal to the floor (axis 2, §4.3); ratio governs the amplitude while absolute size governs the floor (axis 3, §4.5); the residual curves are route-uniform under route-ordered gaps (axis 4, §4.4); and the curve shape follows m/G^P , not per-band SNR (axis 5, §4.4). Taking Eq. 3 with measured $m, G, A_e, \text{Floor}_e$, the account also retrodicts the observations that motivated no probe: (i) $\sigma^* \approx 0.5$ rather than 0.14 — the crossover is where $A_e m/G^P$ sinks below the instrument band, a property of the ratio, so the 3.5× discrepancy is expected, not anomalous; (ii) the 896→768 failure (its floor alone exceeds the band — no σ -gate can rescue it); (iii) 1280→1024 flooring *later* ($\sigma^* \approx 0.75$) but cleanly (harsher ratio $\Rightarrow$ larger A_e ; large target $\Rightarrow$ Floor ≈ 0) — the single observation that breaks both pure-ratio and pure-capacity route-ordering stories.

A corollary worth stating plainly: the safety of 1024→896 is an *empirical smoothness statement about this network’s function across nearby token counts*, not an information-theoretic statement about the data. There was never a reason to expect a universal ratio invariant.

5 FROM THE GAP TO LOW-RESOLUTION TRAINING

5.1 THE (ROUTE, σ) SAFETY MAP

Table 5 is the practical summary of §4: what replaces the spectral rule is not a corrected formula but a measured map. Safety is a per-route boundary $\sigma^*(\text{route})$, jointly determined by ratio (through A_e) and absolute target size (through Floor_e); neither alone predicts the ordering. The batch-aggregate view is, if anything, stronger than the per-image one: summing per-image gradients before the cosine — the object SGD actually follows — collapses the 1024→896 gap to ≈ 0 at every bin $\sigma \geq 0.625$, i.e. the small per-image residual behaves as draw noise that cancels across images rather than as a shared bias. A redundancy-stratified analysis finds no per-image demotion-targeting lever (null trend across strata), consistent with the floor being an image-generic network property.

5.2 σ -CONDITIONAL TRAINING ON THE MEASURED-SAFE ROUTE

We wire the one measured-safe corpus route into the LoRA trainer as an opt-in path, exactly as measured — no extrapolation beyond the map:

σ -first draw. The trainer draws each batch’s σ *before* fetching latents, from the unchanged training density (the σ marginal is untouched — demotion conditions on σ , never the reverse). When every

sample in the batch is above the gate ($\sigma > 0.5$), the native 1024-tier latent is swapped for a cached demoted-grid sibling produced by the probe’s own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream — noise, target, masking, loss — derives from the swapped latent. Off-route images always train native, as does validation.

Banded rotary alignment, σ -gated. Motivated by §4.6, a frequency-selective alignment (YaRN-style bands: full positional-interpolation stretch for rotary bands with few rotations across the demoted extent, native spacing for high-frequency bands, ramp between, with the band thresholds scheduled by a sigmoid gate in σ following Zhao et al., 2026) is applied on demoted forwards only. Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs — it erases the static variant’s low- σ liability ($+0.064 \rightarrow +0.033 \pm 0.025$) while preserving its high- σ gains (best and most stable arm at $\sigma \geq 0.59$ and the endpoint, paired -0.05 vs plain demotion). It is a refinement candidate, not a gate-widener: the gate stays $\sigma > 0.5$.

Cost accounting. With $\sim 50\%$ of draws above the gate and the demoted forward at $\sim 0.72\times$ tokens, the fixed-step epoch cost is $\approx 0.5 \cdot 0.72 + 0.5 = 0.86$ — a $\sim 14\%$ wall-clock ceiling on our corpus (96% of records are on the safe route), scaling with high-tier content. The pitch is wall-clock at fixed steps, never “more steps in the same time.”

Weight-space sanity. Under bit-exact deterministic training (paired runs; without determinism, twin runs put a chaos floor of 0.41 on checkpoint-displacement cosine, which would swamp the read), the rotary refinement adds no displacement over the demotion arm it refines (base $\leftrightarrow$ refined 0.319 vs base $\leftrightarrow$ demoted 0.320), with its footprint concentrated in the low-signal early/mid blocks — consistent with §4.6’s depth profile.

What this section does not claim. The end-to-end acceptance gate — a fixed-step A/B at training scale scoring generation-quality non-inferiority (CMMD; Jayasumana et al., 2024) against the realized wall-clock saving — is designed and pre-committed (a regression closes the line) but not yet run; all claims above are gradient-level and weight-space-level. We flag this deliberately: the gap between gradient-level equivalence at $N=40$ and thousands of optimizer steps is exactly the kind of gap this paper is about.

6 LIMITATIONS

One model family, one operating point. All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint first is the designated bound on how far the map generalizes. The floor’s dependence on the fine-tuning operating point through J is untested (the designated probe: fine-tune at the 1280 tier, re-probe 1280 $\rightarrow$ 1024).

Gradient-level, not end-task. The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed only by the pending A/B above.

The account is a form, not a fit. Eq. 3’s value is that each object is independently measurable and every divergent prediction discriminated cleanly; we fit no coefficients, and the confrontation with the residual-level null (N4) is structural (ordering and floor), not a quantitative fit of its σ -shape. Likewise the band-counting sketch for RoPE_e/Resid_e is calibrated, not derived.

Instrument resolution. Bin-mean SEM ~ 0.02 and the re-encoding band ± 0.04 set the floor on what “safe” can mean; residual gaps of that order are undetectable by construction. The σ^* boundaries are localized only to a bin.

Scope. Adapter (LoRA) gradients on a frozen backbone; full fine-tuning may differ. Pixel-space demotion only (latent-space downsampling was not tested here; SwD found it inferior). The σ -gated banded-alignment parameters were taken from one shot at pre-registered values, not searched.

7 CONCLUSION

“Almost equivalent” input signal is not almost-equivalent training signal. Derived to first order, the training-side cost of demotion has two terms, and the spectral argument — even in its strongest tolerance-parameterized form — is a model of only one of them: a one-parameter, ratio-only family

that cannot express the σ -independent floor the network’s grid-calibrated computation carries. The measurements bear the structure out at every point of divergence: the mildest route’s crossover sits at $3.5\times$ the equal-power prediction and no tolerance reproduces the map; a $2\times$ -downscaled latent never yields the native gradient at any noise level — because expected gradients are not noise-masked, the flow-matching target carries the clean image at every σ , and the network function itself is grid-calibrated in its early blocks, partly through its positional coordinate system and partly through an irreducible dependence of the computation on token count. What replaces the spectral rule is a measured map: per-route boundaries governed by downscale ratio on the noise-dependent term and absolute grid size on the noise-independent one. On the one route our corpus makes safe, the map converts into a free $\sim 14\%$ of wall-clock. We offer the account and its instrument — per- σ -bin gradient agreement with a redraw floor and a re-encoding control, criteria frozen before data — as the cheap, general test that any future “train part of the time at lower resolution” proposal should pass before it is believed.

REPRODUCIBILITY STATEMENT

The probe scripts (`run_sigma_probe.py`, `run_prior_distance.py`, `rapd.py`, `compare_ckpt_dw.py`), the trainer wiring (`--sigma_lowres`), frozen pre-registration documents, and every run’s raw per-image records are in the public repository at https://github.com/sorryhyun/anima_lora (project line `project/sigma_lowres/`). Paired training runs use a `--deterministic` mode (deterministic attention backward included) verified bit-identical across twin runs.

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A INSTRUMENT DETAILS

Estimator. Per image, arm, and σ -bin, adapter gradients are accumulated over D stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at $D=8$ are not comparable *across* bins (the floor drops where $\|g\|$ is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability ≈ 0 and is never used.

Controls. The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; its gap is the validity band (± 0.04 ; observed $|\cdot| \leq 0.054$ across all bins of the main runs, per-module-group ≤ 0.045 in the split runs). The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer’s σ -density reproduces earlier σ -marginalized measurements from the same instrument family (consistency check).

Endpoint and x-zero modes. The $\sigma=1$ endpoint bin feeds input exactly ϵ (the input-information term vanishes by construction). The x-zero mode zeroes x in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the $(1-\sigma)x$ term absent, low- σ bins are off-manifold and the $\sigma=1$ read is primary. In that regime the residual is $\approx -\hat{x}_{\text{prior}}$, so “graph-dominated” includes the model’s grid-conditioned prior; the prior-distance probe (§4.3) subsequently dissociated the prior from the floor’s route ordering.

Spectral-null computation. Table 1 evaluates Eq. 4 at the demoted grids’ Nyquist frequencies $f_{\text{cut}} = \frac{1}{2} e/e_{\text{nat}}$ on the $N=40$ mean radially-averaged latent power spectrum (64 radial bins; P at the 896/768/512 cuts = 0.025/0.029/0.065). The equal-power slice $\delta = (1 + P_{\omega})/2$ recovers $\sigma_{\text{eq}} = \sqrt{P}/(1 + \sqrt{P})$ exactly. Across $\delta \in [0.001, 0.5]$ the mildest–harshes spread never exceeds 0.13, and the 1024→896 versus 896→768 boundaries (cut frequencies 0.4375 vs 0.4286) never separate by more than 0.01. The parameter-free anchor δ_{reenc} of §3.3 is measured by the same estimator: RAPSD of the re-encode arm’s latent error (fresh resize–encode minus cached latent, the gradient probe’s own control chain) on the same probe set, interpolated at each route’s cut frequency [**pending: reenc_noise_floor.py**].

B ADDITIONAL TABLES

Table 6: Ratio-transfer run: 896-tier natives ($N=40$), pre-registered bar “gap₇₆₈ within the re-encoding band at $\sigma \geq 0.5$ ” — **fails** (residual 0.06–0.12, $\sim 2\times$ the 1024→896 plateau).

σ bin	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
896→768	.114	.209	.168	.143	.124	.056	.092	.061
896→512	.318	.372	.308	.340	.320	.280	.329	.217

Table 7: Route-uniformity of the mean prediction residual: cross-grid excess of $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$ over split-half floors (relative L_2 ; same-grid re-encoding control ≤ 0.02 everywhere). The three main routes coincide within $\sim \pm 0.02$ at every σ while their gradient floors span 0 to 0.3.

σ	0.125	0.375	0.625	0.875	1.0
1280→1024	.885	.825	.715	.465	.360
1024→896	.862	.808	.706	.459	.378
896→768	.860	.814	.715	.466	.393
768→512	.897	.872	.767	.508	.435

Table 8: Depth localization of the floor: mean per-block gap at $\sigma=1$, early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry $\sim 3\times$ the late-block gap; the content share (ep – xz) is a late-block minority effect.

route	ep early	ep late	xz early	xz late
1024→512	.357	.223	.351	.125
1024→768	.164	.085	.121	.033
1024→896	.023	.015	.044	.012

Table 9: Positional-interpolation intervention at the $\sigma=1$ endpoint ($N=40$, 16 draws): exact phase-geometry alignment erases the mild-route floor and $\sim 30\%$ of the harsh-route floor.

route	plain (SEM)	PI-aligned (SEM)	paired Δ (SEM)	improved
1024 $\rightarrow$ 896 (control)	-0.021 (.041)	-0.040 (.040)	—	—
1024 $\rightarrow$ 768	+0.080 (.048)	-0.001 (.039)	+0.081 (.031)	78%
1024 $\rightarrow$ 512	+0.320 (.058)	+0.224 (.056)	+0.096 (.039)	70%